ECA1328 – MICROWAVE ENGINEERING

CO1 Application-Based Assessment – Technical Report

Analysis, Design and Verification of a Microwave Transmission Line and Network for a Selected Application

Selected application: Lightweight X-band UAV-borne SAR / weather-radar transmitter-to-antenna feed (9.5–10.5 GHz)

Item	Details
Name / Roll No.	U.Guru Prasad Reddy/192312683L
Course / CO	ECA13 Microwave Engineering – CO1 (K4 Analyze, K5 Evaluate, K6 Create)
Unit covered	Introduction to Microwaves: fundamentals, transmission lines, microwave network analysis
Report flow	APPLICATION → STRUCTURE EVALUATION → MODE MODEL & DESIGN → LOSS EVALUATION → NETWORK / S-PARAMETER MODEL → VALIDATION → ENGINEERING DECISION
Selected structure	Air-filled rectangular waveguide WR-90, dominant TE10 mode (aluminium, silver-plated, polished bore)
Final decision	Initial design: NO. Final design: YES (with a ±2 % machining-tolerance class)

Consolidated Question :

Select a practical microwave application (civil, military, medical, industrial or EMI/EMC) and formulate a quantified specification of impedance, frequency band, insertion loss, return loss and isolation for its transmission structure. Evaluate the coaxial line, waveguide, stripline and microstrip line in terms of their modes (TEM, quasi-TEM, TE and TM), cutoff behaviour, dispersion and losses, and select one with a weighted justification; derive the transmission-line and propagation model and design the dimensions so that only the intended mode propagates; develop a loss budget, identify the dominant loss and design a loss-reduction measure; create an equivalent voltage–current and Z/Y/ABCD network model, derive the S-parameters, design a matching arrangement and evaluate reciprocity, losslessness, return loss and insertion loss; validate the design with a tolerance analysis, EMC consideration and specification verification table; and decide, with quantitative justification, whether the structure and mode can be adopted.

Summary of the result

Design in one line: a 0.5 m WR-90 (22.86 mm × 10.16 mm) air-filled waveguide carrying only the TE10 mode from 9.5 to 10.5 GHz (cutoff 6.557 GHz; next mode TE20 at 13.114 GHz), fed from the transmitter through a coax-to-waveguide transition and ended on a reduced-height antenna feed by an E-plane λg/4 transformer (7.87 mm high, 9.93 mm long).

Key figure (worst case in 9.5–10.5 GHz)	Specified	Initial design	Final design
Total insertion loss (0.5 m run)	≤ 0.30 dB	0.55 dB ✗	0.16 dB nominal, 0.23 dB at ±5 % tolerance ✓

Key figure (worst case in 9.5–10.5 GHz)	Specified	Initial design	Final design
Return loss / VSWR	≥ 20 dB / ≤ 1.22	12.2 dB / 1.65 ✗	29.0 dB / 1.07 nominal; 24.1 dB / 1.13 at ±2 % ✓
Single-mode margin	≥ 25 % above fc10, ≥ 15 % below fc20	45 % / 20 % ✓	45 % / 20 % nominal; 38 % / 16 % at ±5 % ✓
Peak power breakdown (sea level)	≥ 500 W + 10 dB	≈ 1.05 MW ✓	≈ 1.05 MW ✓

Note on numbers. All propagation, loss and S-parameter values below are computed from closed-form models (code in Appendix A) and checked by two independent routes. Connector/transition and flange losses and the EMI shielding of the flange are design allowances taken from typical practice; they are marked "allowance" and must be confirmed by measurement (Stage 6 test plan). A full-wave CST/HFSS check of the steps and the transition is listed as the next verification step; it was not run for this report.

STAGE 1 – Formulate the Application and Operating Requirement

1.1 Application, category and importance

Application: the transmitter-to-antenna feed of a lightweight X-band synthetic-aperture radar (SAR) / weather radar carried by a medium-size UAV. A solid-state power amplifier delivers 500 W peak (10 % duty, 50 W average) and the energy must reach the antenna feed 0.5 m away with minimal loss, a clean single mode and strong isolation from the UAV's other radios and avionics.

Category: civil / military dual use (disaster mapping, flood and crop monitoring, border and maritime surveillance, weather sensing). Every 0.1 dB lost in the feed is radiated power lost on both the transmit and receive paths (two-way radar equation), and the feed also sits next to GNSS, C2 datalink and Wi-Fi receivers, so shielding and mode purity matter as much as loss.

1.2 Historical development (why the choice is natural)

- **1865–1887:** Maxwell predicts electromagnetic waves; Hertz generates and detects them at about 0.5–1 m wavelength, the first microwave-scale experiments.
- **1897–1936:** Rayleigh derives the modes of hollow metal pipes; Southworth and Barrow (Bell Labs and MIT, 1936) demonstrate practical waveguide transmission – the origin of the TE/TM mode concept used here.
- **1940–45:** the cavity magnetron (Randall and Boot, Birmingham) makes compact high-power microwave radar possible; waveguide becomes the standard high-power feed.
- **1950s–70s:** coaxial, stripline and microstrip lines enable airborne and satellite links; Telstar-class satellite communication begins in the early 1960s.
- **1980s–today:** GaAs/GaN MMICs, phased arrays, automotive radar, 5G/6G mmWave. Planar lines dominate low-power chips, but high-power, low-loss feeds still rely on waveguide.

1.3 Frequency-band comparison

Band	Frequency	Wavelength λ0	Rain attenuation (≈25 mm/h, order of magnitude)	Component size / remark
C	4–8 GHz (use 6 GHz)	50 mm	≈ 0.1 dB/km	WR-137 guide a = 34.8 mm; bulky; coarse resolution (B ≈ 0.5 GHz)
X (selected)	8–12 GHz (use 10 GHz)	30 mm	≈ 0.7 dB/km	WR-90 guide a = 22.9 mm; B up to ≥1 GHz; antenna ≈ 30 cm for ≈ 7° beam
Ku	12–18 GHz (use 15 GHz)	20 mm	≈ 1.5–2 dB/km	WR-62 guide a = 15.8 mm; smaller, but rain loss and PA cost rise

Table 1: Band trade-off (rain figures follow the ITU-R P.838 trend and are indicative only).

Decision: X-band, centre frequency 10 GHz, operating band 9.5–10.5 GHz (B = 1 GHz, 10 % fractional). It gives 0.15 m range resolution (c/2B), a compact 22.9 mm guide, modest rain loss and mature, inexpensive transmitter components; C-band is too bulky and has half the bandwidth, Ku has higher rain loss and tighter machining tolerances.

1.4 Quantified requirement specification

Parameter	Requirement	Basis
Application / category	X-band UAV SAR / weather-radar feed; civil + military	Stage 1.1
Operating frequency / band	10 GHz centre; 9.5–10.5 GHz; length 0.5 m	Range resolution, size
Characteristic impedance	50 Ω at the coaxial transmitter port; guide equivalent impedance Zpv = 443.5 Ω (TE10, referred to the power–voltage definition)	Standard coax interface

Parameter	Requirement	Basis
Maximum insertion loss	≤ 0.30 dB over the 0.5 m run (all mechanisms incl. transition, flanges, mismatch)	≈ 7 % power budget
Input return loss / VSWR	≥ 20 dB (VSWR ≤ 1.22) across 9.5–10.5 GHz	Protects PA from reflection
Isolation / EMI shielding	≥ 90 dB shielding effectiveness; out-of-band (< 7 GHz) rejection ≥ 60 dB	GNSS / C2 / Wi-Fi co-site
Power level	500 W peak, 50 W average; breakdown margin ≥ 10 dB at 5 km altitude	PA rating
Size / weight / cost	Mass ≤ 250 g incl. flanges; cross-section ≤ 30 × 15 mm; unit cost target ≤ US$ 400	UAV payload
Mode purity	Only one mode over 9.5–10.5 GHz with ≥ 25 % margin above the dominant cutoff and ≥ 15 % below the next mode	Phase-stable, repeatable feed

Table 2: Application specification.

1.5 Constraints

- **EMI/EMC:** the feed is a potential radiator of 500 W peak; a closed metal guide with a gasketed or choke flange keeps leakage ≥ 90 dB below carrier. This favours a closed structure over open planar lines.
- **Safety (power density):** with leakage ≥ 90 dB the feed leaks < 0.5 µW peak, far below any exposure limit; the real exposure zone is in front of the antenna and is controlled operationally.
- **Environment:** −40 to +60 °C, vibration and pressure change with altitude (air breakdown strength falls with pressure), humidity (silver tarnish). These limit dielectric-filled structures and favour a rigid, sealed, air-filled guide.

Stage 1 deliverable – Requirement Specification: Application (X-band UAV radar feed) → Category (civil/military) → Band 9.5–10.5 GHz (justified in Table 1) → Transmission requirements (Table 2) → Constraints (1.5) → Acceptance criteria: IL ≤ 0.30 dB, RL ≥ 20 dB, shielding ≥ 90 dB, single-mode margins ≥ 25 % / ≥ 15 %, mass ≤ 250 g.

STAGE 2 – Evaluate and Select the Transmission Structure

2.1 Modes and field structure at 10 GHz

Structure	Wave type	Dominant mode / next mode	Cutoff	Dispersion	Zc and loss at 10 GHz
Coax (RG-402 type: a = 0.455 mm, b = 1.49 mm, PTFE εr = 2.1)	TEM	TEM (no cutoff) / TE11 at 33.9 GHz	0 (DC) for TEM	None: vp = c/√εr = 2.07×10^8 m/s	Zc = 49.1 Ω; αc 1.05 + αd 0.26 = 1.32 dB/m
Rectangular waveguide WR-90 (air)	TE (and TM)	TE10 / TE20 at 13.11 GHz	fc10 = 6.557 GHz	Strong: vp > c, vg < c	Z TE10 = 499 Ω; 0.12–0.30 dB/m
Stripline (b = 1.0 mm, w = 0.54 mm, εr = 3.66, tanδ 0.0037)	TEM	TEM / parallel-plate modes (avoid by b < λ/(2√εr))	0 for TEM	None (homogeneous)	Zc = 50 Ω; αc 4.4 + αd 6.4 = 10.9 dB/m
Microstrip (h = 0.508 mm, w = 1.11 mm, RO4350B)	Quasi-TEM	Quasi-TEM / surface-wave TE1, TM0 (TM0 has no cutoff)	0 (quasi-TEM)	Mild: εeff 2.85 → 2.90 at 10 GHz	Zc = 50.2 Ω; αc 6.1 + αd 5.1 ≈ 11.2 dB/m + radiation

Table 3: Mode and loss comparison (first-order closed-form estimates, copper conductors; coax and planar lines scaled from standard formulas).

Why microstrip is quasi-TEM and not pure TEM. A pure TEM wave requires every field line to travel at the same phase velocity in all media. In microstrip, part of the field lies in air (vp = c) and part in the substrate (vp = c/√εr). Continuity of the tangential fields at the air–dielectric interface forces one common β, so the wave cannot be TEM in both regions; the true mode is a hybrid with small

longitudinal Ez and Hz components. Because h ≪ λ (0.508 mm vs 30 mm) these components are tiny and the static (quasi-TEM) approximation with an effective permittivity εeff = (εr+1)/2 + (εr−1)/2·(1+12h/w)^(-1/2) = 2.85 works well; the weak frequency dependence of εeff is the dispersion.

Why a hollow waveguide cannot support TEM. A TEM wave needs Ez = Hz = 0, which makes the transverse field the gradient of a potential Φ that satisfies Laplace's equation ($\nabla_t^2 \Phi = 0$). A hollow guide has a single conductor, so Φ is constant on its boundary; the only solution of Laplace's equation with a constant boundary value is Φ = constant, i.e. zero field. Hence only TE and TM modes (with a cutoff frequency) exist.

2.2 Weighted decision matrix

Scores run from 1 (poor) to 5 (best) and are anchored on the numbers of Table 3: loss score from dB over 0.5 m (waveguide ≈ 0.06 dB, coax ≈ 0.66 dB, planar ≈ 5.5 dB); power from breakdown/thermal limits (waveguide ≈ 1 MW, coax ≈ 27 kW peak, thermal limit of tens of W average, planar lines far lower); shielding from closed-wall construction.

Criterion	Weight	Waveguide (A)	Coax (B)	Stripline (C)	Microstrip (D)	Remarks
Insertion loss over 0.5 m	25 %	5	3	2	1	dB: 0.06 / 0.66 / 5.4 / 5.6
Power handling	20 %	5	3	2	1	≈1 MW / 27 kW
EMI shielding / isolation	15 %	5	4	4	2	closed vs open
Frequency range / single-mode bandwidth	10 %	4	5	4	3	WG ≈ 40 % band
Size and weight	10 %	2	4	4	5	WG bulkiest
Cost	10 %	2	3	3	5	machining
Integration / manufacturability	10 %	2	3	4	5	planar on PCB
Overall weighted score (max 5)	**100 %**	**4.00**	**3.45**	**3.00**	**2.55**	**Selected: A**

Table 4: Weighted selection table (criterion, weight, structure A, B, C, D, selected structure, overall score).

2.3 Sensitivity analysis

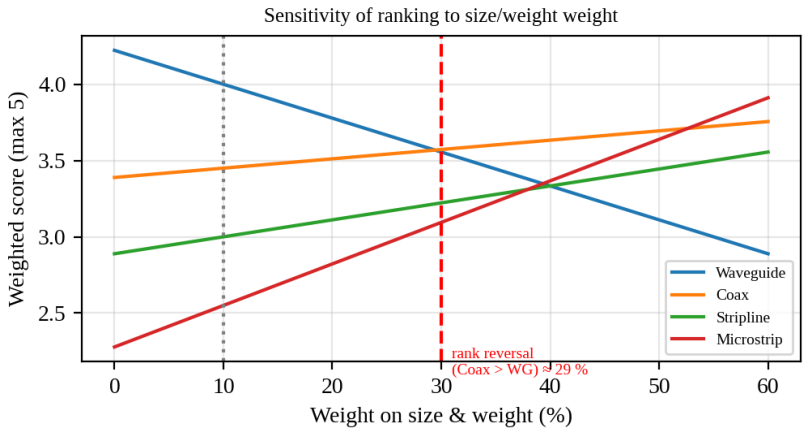


Figure 1: Overall score versus the weight on size and weight (other weights rescaled proportionally).

- **Weights:** raising the weight of size/weight from 10 % to about **29 %** makes coax overtake waveguide (A: 4.22(1−s)+2s; B: 3.39(1−s)+4s are equal at s ≈ 0.29). Equivalently, if size + cost + integration together exceed about **51 %** of the total weight (baseline 30 %), coax wins. Microstrip never wins at any realistic weighting because its loss and power scores are 1.

- **Frequency ×2 (20 GHz):** WR-90 becomes multimode (TE20 13.11, TE01 14.75, TE11/TM11 16.15, TE30 19.67 GHz all propagate), so the guide must be re-sized to WR-42 (a = 10.67 mm); waveguide stays lowest-loss (≈ 0.39 dB/m copper) and its size penalty shrinks. Coax loss rises to ≈ 2.0 dB/m (TE11 cutoff 33.9 GHz is still safe) and the planar lines get worse (dielectric loss grows ∝ f). The ranking therefore does not reverse; waveguide's lead grows.
- **Frequency ÷2 (5 GHz):** WR-90 is below cutoff (evanescent attenuation ≈ 772 dB/m); a 47.5 mm WR-187 (fc = 3.15 GHz, ≈ 0.03 dB/m) is needed – twice as wide, roughly 4× the cross-section. Size and weight dominate and the **ranking reverses towards coax** (≈ 0.88 dB/m, 0.44 dB per 0.5 m) at around 5 GHz or lower for a 0.5 m run. The un-selected coax is therefore the right answer for L/S-band or very short, low-power links.

2.4 Defence against the strongest alternative (coax)

Semi-rigid coax is lighter, cheaper and has no cutoff, but over 0.5 m it loses 0.66 dB (13.8 % of power, 7 W heat at 50 W average) against 0.06 dB for the silver-plated guide, so the coax loss alone is more than twice the whole 0.30 dB budget. Its power handling is limited to tens of watts average by dielectric heating (the 27 kW figure is only the air-breakdown limit at the centre conductor), and its centre conductor (0.9 mm) risks multipactor/arcing at altitude. Waveguide costs roughly 0.15–0.2 kg more mass (estimate) to save ≈ 0.6 dB and gain 90 dB-class shielding.

> **Design Decision 1:** For the specified application I select **WR-90 rectangular waveguide (TE10)** over **semi-rigid coaxial line** because it meets the 0.30 dB loss budget with margin (≈ 0.06 dB vs 0.66 dB over 0.5 m), has ≈ 1 MW breakdown capability against ≈ 27 kW (tens of watts thermal) for coax, and is a closed shield. If the frequency were halved the guide would grow to 47.5 mm wide and the coax would win on size; if doubled, WR-90 turns multimode and must be replaced by WR-42, while coax loss rises to ≈ 2 dB/m, so for the structure not selected (coax) the penalty grows with frequency.

STAGE 3 – Develop the Mode and Propagation Model

3.1 Transmission-line model (telegrapher's equations)

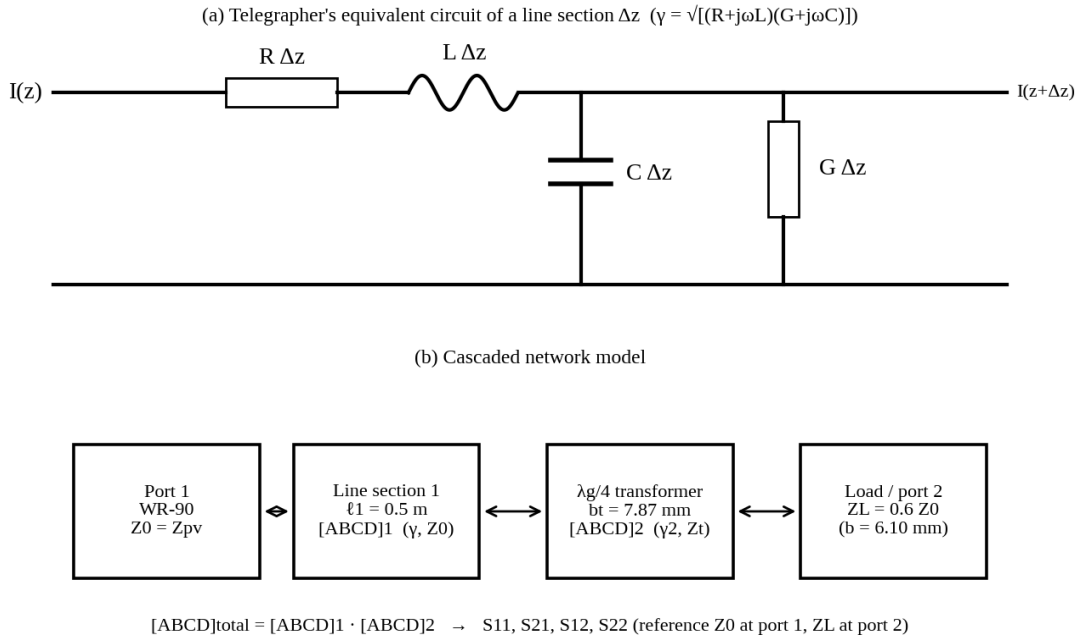


Figure 2: (a) Distributed-element model of a line section; (b) cascaded ABCD network of the designed feed.

For a short section Δz with series R, L and shunt G, C per unit length, Kirchhoff's laws give, for e^{jωt} time dependence:

$$dV/dz = -(R + j\omega L)\, I\,, \qquad dI/dz = -(G + j\omega C)\, V$$

Eliminating I gives the wave equation $d^2V/dz^2 = \gamma^2 V$ with

$$\gamma = \alpha + j\beta = \sqrt{[(R + j\omega L)(G + j\omega C)]}\,, \qquad V(z) = V^{+}e^{(-\gamma z)} + V^{-}e^{(+\gamma z)}$$

Low-loss approximation ($R \ll \omega L$, $G \ll \omega C$):

$$\beta \approx \omega\sqrt{(LC)}\,, \quad \alpha \approx R/(2Z0) + G\cdot Z0/2 = \alpha c + \alpha d$$

$$Z0 = \sqrt{[(R+j\omega L)/(G+j\omega C)]} \approx \sqrt{(L/C)}$$

$$\Gamma(z=0) = (ZL - Z0)/(ZL + Z0)\,, \quad Zin(\ell) = Z0\,(ZL + Z0 \tanh \gamma\ell)/(Z0 + ZL \tanh \gamma\ell)$$

3.2 Field solution for the rectangular guide

Assumptions: (i) perfectly conducting walls for the field solution (losses are later found by perturbation, valid because the skin depth $\delta \approx 0.7$–1 µm $\ll b$); (ii) homogeneous, lossless, air-filled guide (ε0, µ0); (iii) uniform cross-section along z; (iv) time-harmonic fields, propagation e^(−jβz).

Maxwell's curl equations give the Helmholtz equation $(\nabla^2 + k^2)H = 0$ with $k = \omega\sqrt{(\mu 0 \varepsilon 0)}$. For **TE modes** ($E_z = 0$) write $H_z = hz(x,y)e^{(-j\beta z)}$:

$$\partial^2 hz/\partial x^2 + \partial^2 hz/\partial y^2 + kc^2\, hz = 0\,, \quad kc^2 = k^2 - \beta^2$$

Separation of variables $hz = X(x)Y(y)$ gives $X'' = -kx^2 X$ and $Y'' = -ky^2 Y$. The boundary condition on a perfect conductor ($\partial hz/\partial n = 0$ on $x = 0$, a and $y = 0$, b) leaves

$$hz = A_mn \cos(m\pi x/a) \cos(n\pi y/b)\,, \quad kc = \sqrt{[(m\pi/a)^2 + (n\pi/b)^2]}\,, \quad \beta = \sqrt{(k^2 - kc^2)}$$

For **TM modes** ($H_z = 0$) the condition $E_z = 0$ on the walls gives $ez = B_mn \sin(m\pi x/a) \sin(n\pi y/b)$ with $m, n \geq 1$. The cutoff frequency ($\beta = 0$) is $fc,mn = (c/2)\sqrt{[(m/a)^2 + (n/b)^2]}$. For the dominant TE10 mode ($m = 1$, $n = 0$):

$$Hz = A10 \cos(\pi x/a)\, e^{(-j\beta z)}$$

$$Hx = (j\beta a/\pi)\, A10 \sin(\pi x/a)\, e^{(-j\beta z)}\,, \quad Ey = -(j\omega\mu a/\pi)\, A10 \sin(\pi x/a)\, e^{(-j\beta z)}$$

$$Z_TE = Ey/(-Hx) = \omega\mu/\beta = \eta 0/\sqrt{(1-(fc/f)^2)}\,, \quad Z_TM = \eta\beta/k = \eta 0\sqrt{(1-(fc/f)^2)}$$

3.3 Mode comparison (TEM, TE, TM) in this guide

Mode	Field components	Cutoff behaviour	Dispersion	Wave impedance
TEM	Ez = Hz = 0	Would have no cutoff, but cannot exist (Φ ≡ const)	n/a	n/a
TE10 (dominant)	Ey, Hx, Hz; one half-sine variation in x	fc = c/2a = 6.557 GHz (lowest of all modes)	Strong: vp = c/√(1−x²), vg = c√(1−x²), vp·vg = c²	Z_TE = η0/√(1−x²) = 499 Ω at 10 GHz
TE20	Ey, Hx, Hz; full sine in x	13.114 GHz	as above	→ η0 as f ≫ fc
TM11 (lowest TM)	Ez, Ex, Ey, Hx, Hy	16.145 GHz (degenerate with TE11)	as above	Z_TM = η0√(1−(fc/f)²) – evanescent at 10 GHz

Table 5: Mode comparison (x = fc10/f).

Best mode: TE10 – lowest cutoff, simplest field (single E-component, polarisation fixed), widest single-mode band (an octave from 6.56 to 13.11 GHz), lowest conductor loss, and it matches the linear polarisation of a slotted-array/horn antenna. TM modes need both m, n ≥ 1, so their lowest cutoff is 16.1 GHz, well outside the band.

3.4 Dimension design

Design rules for single-mode operation over 9.5–10.5 GHz: (1) fc10 ≤ 0.75 fmin → a ≥ c/(2·0.75·9.5 GHz) = 21.0 mm; (2) fmax ≤ 0.85 fc20 (≥ 15 % margin) → a ≤ c/(10.5/0.85 GHz) = 24.3 mm; (3) b ≤ a/2 so that TE01 (fc = c/2b) stays above TE20; also b large enough for power. Choosing the standard WR-90 values satisfies all three:

$$a = 22.86 \text{ mm } (0.900 \text{ in}), \quad b = 10.16 \text{ mm } (0.400 \text{ in}) = a/2.25$$

Worked calculation at f = 10 GHz (units shown)

$$fc10 = c/(2a) = 2.998\times10^{8} / (2 \times 22.86\times10^{-3}) = 6.557 \text{ GHz}$$

$$x = fc/f = 0.6557\,, \quad \sqrt{(1-x^2)} = 0.7550$$

$$k0 = 2\pi f/c = 209.6 \text{ rad/m}\,, \quad kc = \pi/a = 137.4 \text{ rad/m}$$

$$\beta = \sqrt{(k0^2 - kc^2)} = \sqrt{(43\,924 - 18\,887)} = 158.2 \text{ rad/m}$$

$$\lambda g = 2\pi/\beta = 39.71 \text{ mm } (\lambda 0 = 30.0 \text{ mm})$$

$$vp = \omega/\beta = 3.971\times10^{8} \text{ m/s} = 1.324\ c\,, \quad vg = c\sqrt{(1-x^2)} = 2.263\times10^{8} \text{ m/s} = 0.755\ c$$

$$\text{check: } vp\cdot vg = 8.99\times10^{16} \text{ m}^2/\text{s}^2 = c^2\ \checkmark\,, \quad \text{group delay } \tau g = L/vg = 0.5 / 2.263\times10^{8} = 2.21 \text{ ns}$$

$$Z_TE10 = \eta 0/\sqrt{(1-x^2)} = 376.73/0.7550 = 499.0\ \Omega$$

Parameter	9.5 GHz	10.0 GHz	10.5 GHz	Comment
fc10 (GHz)	6.557	6.557	6.557	dominant cutoff
fc/f	0.690	0.656	0.624	
β (rad/m)	144.1	158.2	171.9	
λg (mm)	43.61	39.71	36.56	λ0 = 31.6/30.0/28.6 mm
vp (10^8 m/s)	4.143	3.971	3.838	> c (no energy transport)
vg (10^8 m/s)	2.169	2.263	2.341	< c
Group delay over 0.5 m (ns)	2.305	2.209	2.136	Δτ = 0.17 ns across band
Z_TE10 (Ω)	520.6	499.0	482.3	equivalent Zpv = 2b/a·Z_TE = 443.5 Ω at 10 GHz

Table 6: Designed dimensions and propagation parameters (a = 22.86 mm, b = 10.16 mm, L = 0.5 m).

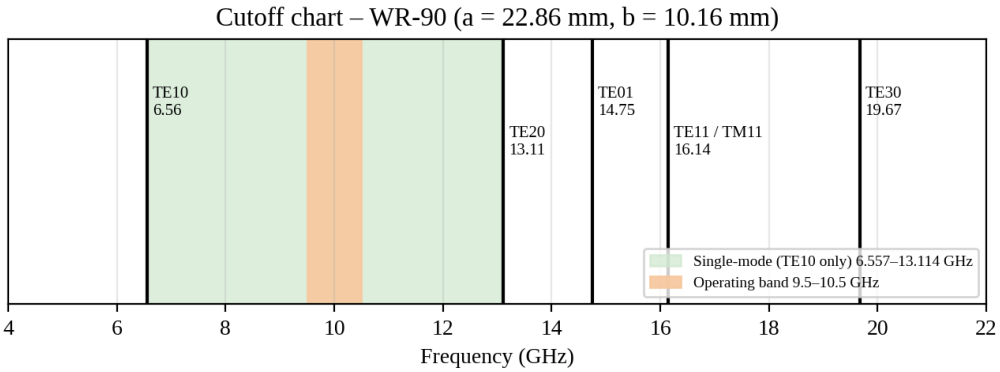


Figure 3: Cutoff chart with the operating band and the single-mode region.

Mode cutoffs (GHz)	Value	Margin to the band
TE10 (dominant)	6.557	9.5/6.557 = 1.45 → 44.9 % above cutoff ✓
TE20	13.114	10.5/13.114 = 0.80 → 19.9 % below next mode ✓
TE01	14.754	29 % margin
TE11 / TM11 (degenerate pair)	16.145	35 % margin
TE30	19.671	—

Power handling check: $P_{breakdown} = E_{br}^2 \cdot a \cdot b / (4 Z_{TE})$ with $E_{br} = 3$ MV/m (air, sea level) = $(3\times10^6)^2 \times 22.86\times10^{-3} \times 10.16\times10^{-3} / (4 \times 499)$ = **1.05 MW**, i.e. 33 dB above the 500 W peak. Even with a derating of 10 for altitude, VSWR and a 3 dB safety factor, margin exceeds 20 dB.

3.5 Higher-order mode risk and design changes

- **Mode degeneracy:** TE11 and TM11 share the same cutoff, 16.145 GHz (a degenerate pair that can convert into each other at asymmetric discontinuities). Both are 35 % above the band, so they matter only for the second harmonic (20 GHz) of a non-linear PA.
- **Feed/transition effect:** a coax probe or a step excites any mode whose field is not orthogonal to the discontinuity. TE20 is odd about x = a/2, so a probe on the centre line (x = a/2) excites no TE20 even if it were above cutoff; an off-centre or tilted transition would. TE20 is also 2.6 GHz above the band so it is evanescent anyway (attenuation $\approx (2\pi/c)\sqrt{(fc20^2 - f^2)} \approx 165$ Np/m at 10.5 GHz).
- **Design change:** keep the probe on the centre line with a symmetric backshort, keep the E-plane steps symmetrical about the guide axis, and put a 2nd-harmonic (≥ 40 dB at 20 GHz) waveguide low-pass filter or absorber vane at the PA output; add polished, flush flanges to avoid asymmetric gaps that couple to TE20/TE11.

> **Design Decision 2:** The dominant mode is **TE10** with cutoff frequency **6.557 GHz** (next mode TE20 at **13.114 GHz**); the single-mode range is **6.557 GHz** to **13.114 GHz**, which contains 9.5–10.5 GHz with margins of 44.9 % above and 19.9 % below; therefore the chosen dimensions (a = 22.86 mm, b = 10.16 mm) **are acceptable** because only TE10 propagates, the dispersion is moderate (group delay change 0.17 ns across the band) and 1.05 MW breakdown power is far above 500 W.

STAGE 4 – Evaluate Losses and Design a Loss-Reduction Strategy

4.1 Loss mechanisms and formulas

$$R_s = \sqrt{(\pi f \mu_0/\sigma)}\ , \quad \delta_s = 1/\sqrt{(\pi f \mu_0 \sigma)}$$

$$K_{rough} = 1 + (2/\pi)\arctan[1.4\,(\Delta/\delta_s)^2] \quad \text{(Hammerstad)}$$

$$\alpha_c\,(TE10) = K_{rough}\,R_s / [\,b\,\eta_0\,\sqrt{(1-x^2)}\,] \cdot [\,1 + (2b/a)\,x^2\,] \quad \text{Np/m}$$

$$\alpha_d = k^2 \tan\delta / (2\beta) \quad \text{Np/m}$$

Radiation is zero for the closed guide except for leakage through flange seams, and mismatch loss is $ML = -10\log(1 - |\Gamma|^2)$. Connector/transition values are allowances (to be confirmed by measurement). 1 Np = 8.686 dB.

Worked calculation – conductor attenuation, initial design (6061 aluminium, $\sigma \approx 2.5\times10^7$ S/m, 1.5 µm rms bore)

$$R_s = \sqrt{(\pi \times 10^{10} \times 4\pi\times10^{-7} / 2.5\times10^7)} = 0.0397\ \Omega\ ; \quad \delta_s = 1.007\ \mu m\ ; \quad \Delta/\delta_s = 1.49$$

$$K_{rough} = 1 + 0.6366 \times \arctan(1.4 \times 1.49^2) = 1 + 0.6366 \times 1.259 = 1.802\ ; \quad R_{s,eff} = 0.0716\ \Omega$$

$$b\,\eta_0\,\sqrt{(1-x^2)} = 0.01016 \times 376.73 \times 0.7550 = 2.890\ \Omega\cdot m\ ; \quad 1 + (2b/a)x^2 = 1 + 0.8889 \times 0.4300 = 1.382$$

$$\alpha_c = 0.0716 / 2.890 \times 1.382 = 0.0342\ \text{Np/m} = 0.297\ \text{dB/m} \rightarrow 0.149\ \text{dB over 0.5 m}$$

Worked calculation – dielectric attenuation

$$\text{Air: } \tan\delta \approx 3\times10^{-8} \rightarrow \alpha_d = (209.6)^2 \times 3\times10^{-8} / (2 \times 158.2) = 4.2\times10^{-6}\ \text{Np/m} = 3.6\times10^{-5}\ \text{dB/m}\ \text{(negligible)}$$

For comparison, a PTFE-filled guide ($\tan\delta = 2\times10^{-4}$, $\varepsilon_r = 2.1$) scaled to keep the same cutoff would have $\alpha_d = k^2\tan\delta/2\beta \approx 0.35$ dB/m – almost three times the conductor loss of the improved guide. If the width a were kept instead, TE20 would drop to 9.05 GHz, below the band, and the guide would be multimode. Dielectric loading is therefore rejected.

4.2 Loss budget and dominant loss

Loss component	Initial (dB/m)	Improved (dB/m)	% of total (initial)	Initial over 0.5 m (dB)	Improved over 0.5 m (dB)	Trade-off / remarks
Conductor loss	0.2975	0.1280	27.0 %	0.149	0.064	Silver plating adds cost, tarnish risk; polishing adds cost
Dielectric loss	0.00004	0.00004	≈ 0	0.00002	0.00002	Air-filled; no dielectric support
Radiation / flange leakage	0.0040	0.0010	0.4 %	0.002	0.0005	Choke / gasketed flange (allowance)
Connector / transition	0.2400	0.1800	21.8 %	0.120	0.090	Tuned coax-to-WG probe + 2 flanges (allowance)
Mismatch (reflection)	0.5606	0.0110	50.9 %	0.280	0.0055	λg/4 transformer; worst case in band
Total attenuation	**1.1021**	**0.3201**	**100 %**	**0.551**	**0.160**	**Limit 0.30 dB**

Table 7: Loss table (initial: aluminium, 1.5 µm rms, direct step to the load; improved: silver-plated, 0.3 µm rms, λg/4 transformer).

Dominant loss. In the initial design the largest contribution is **mismatch (reflection) loss, 0.28 dB = 50.9 %** of the 0.55 dB total, caused by the 12 dB return loss of the unmatched load. Among the genuinely dissipative mechanisms, conductor loss dominates (0.149 of 0.271 dB, 55 %), followed by the transition (0.12 dB, 44 %); radiation and dielectric are negligible. After improvement the transition/connector becomes the largest term (56 % of 0.16 dB), conductor loss is 40 % and mismatch only 3.4 %.

4.3 Parametric study

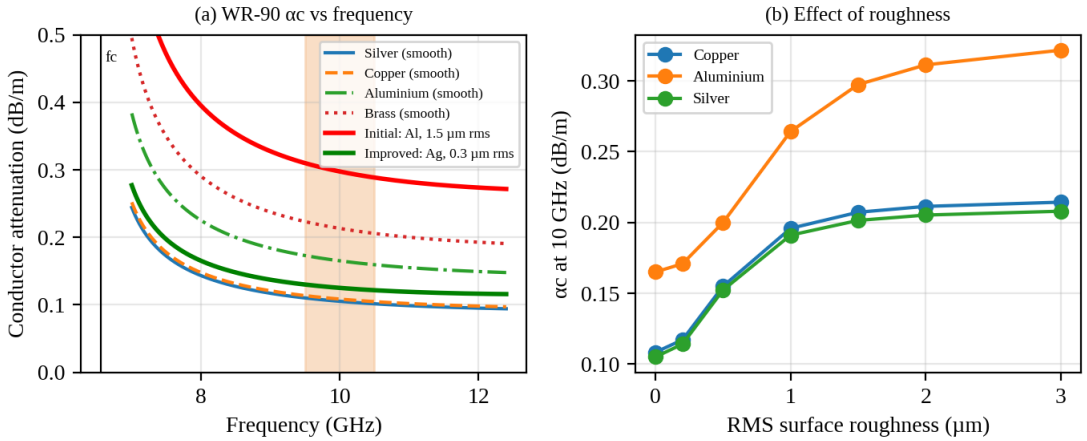


Figure 4: (a) Conductor attenuation of WR-90 versus frequency for different conductors (band shaded); (b) effect of surface roughness at 10 GHz.

Conductor (smooth wall, 10 GHz)	Silver	Copper	Gold	Aluminium	Brass
σ (10^7 S/m)	6.17	5.80	4.10	2.50	1.50
Skin depth δs (µm)	0.64	0.66	0.79	1.01	1.30
Rs (Ω)	0.0253	0.0261	0.0310	0.0397	0.0513
αc (dB/m)	0.105	0.108	0.129	0.165	0.213

Roughness rms (µm)	0	0.2	0.5	1.0	1.5	3.0
Copper αc (dB/m)	0.108	0.117	0.155	0.196	0.207	0.214
Aluminium αc (dB/m)	0.165	0.171	0.200	0.264	0.297	0.322
Silver αc (dB/m)	0.105	0.114	0.152	0.191	0.202	0.208

Dielectric-filled guide (hypothetical), 10 GHz	tanδ = 0 (air)	1×10^{-4}	2×10^{-4} (PTFE)	1×10^{-3}
αd (dB/m)	≈ 0	0.175	0.349	1.747

Interpretation. (i) αc varies slowly over the band (9.5 GHz: 0.310, 10 GHz: 0.297, 10.5 GHz: 0.288 dB/m for the initial guide) and decreases as f moves away from cutoff, reaching a shallow minimum above the band, so a single-mode band loses little at either edge. (ii) Material matters through Rs ∝ 1/√σ: switching from aluminium to silver saves 36 % (0.165 → 0.105 dB/m). (iii) Roughness is as important as the metal: 1.5 µm rms on aluminium raises loss by 80 % because the roughness factor saturates at 2 once Δ > 2δs, so a 0.3 µm polished bore is needed with a metal of δs ≈ 0.65 µm. (iv) Dielectric loss scales linearly with tanδ and, if any dielectric were used, would exceed conductor loss; the air fill removes it.

4.4 Redesign and judgement

1. **Silver plating (≥ 5 µm ≈ 8δs) with polished bore (0.3 µm rms):** conductor loss 0.297 → 0.125 dB/m, i.e. 0.149 → 0.063 dB over 0.5 m (−0.086 dB).
2. **λg/4 E-plane transformer** (Stage 5): mismatch loss 0.280 → 0.0055 dB worst case in band (−0.275 dB).
3. **Optimised coax-to-waveguide probe and gasketed/choke flange** (allowance): transition + flanges 0.12 → 0.09 dB, leakage 0.002 → 0.0005 dB.

Total attenuation falls from **0.551 dB to 0.160 dB (−0.39 dB, −71 %)**. **Judgement:** the improved design meets the Stage 1 limit (0.30 dB) with 0.14 dB margin and still meets it at ±5 % tolerance (0.23 dB,

Stage 6). **Trade-offs:** plating cost and silver tarnish (a 0.1 μm gold flash or anti-tarnish lacquer thinner than δs is needed), tighter bore tolerance and polishing cost, and the transformer's bandwidth is finite (RL ≥ 20 dB still holds from about 8.6 to 11.5 GHz, Stage 5). The transition is now the largest loss (56 %), which defines the next improvement.

> **Design Decision 3:** The dominant loss mechanism is **mismatch (reflection) loss**, contributing **50.9 %** of the total (conductor loss 27.0 % and the dominant dissipative loss); the loss-reduction measure I adopt is **silver plating + polished bore, a λg/4 matching transformer and an optimised transition/flange**, which reduces the attenuation from **0.551 dB (1.10 dB/m)** to **0.160 dB (0.32 dB/m)**.

STAGE 5 – Create and Evaluate the Network Model

5.1 Equivalent voltage and current for a non-TEM line

In a TEM line the transverse field is the gradient of a static potential, so V = −∫E·dl and I = ∮H·dl are unique and independent of the path. In a waveguide the field is TE/TM, ∮E·dl depends on the path and the line integral of H depends on where it is taken, so **no unique V and I exist**. Equivalent V and I must be defined so that (i) V ∝ transverse E, I ∝ transverse H, and (ii) the complex power P = ½VI* equals the power in the field. For TE10 with Ey = E0 sin(πx/a)e^(−jβz) and Hx = −Ey/Z_TE, the power is P = ab E0²/(4 Z_TE) and three common choices give three different impedances:

Definition	Z0 expression	Value at 10 GHz
Power–voltage (V = E0 b, P)	Z_PV = V²/(2P) = (2b/a) Z_TE	443.5 Ω
Voltage–current (V = E0 b, I = ∫Hx dx = 2aE0/(πZ_TE))	Z_VI = V/I = (πb/2a) Z_TE	348.3 Ω
Power–current (I, P)	Z_PI = 2P/I² = (π²b/8a) Z_TE	273.6 Ω

The three values differ by up to 62 %, which is the ambiguity. S-parameters are unaffected provided one definition is used consistently, because only impedance **ratios** matter (all three are proportional to b/a·Z_TE). I adopt **Z0 = Z_PV = (2b/a) Z_TE10 = 443.5 Ω** (power-normalised) and note that a height change from b to b' changes Z0 in the ratio b'/b (the principle behind the E-plane transformer). Wave variables referred to a real Z0 are:

$$a = (V + Z_0 I) / (2\sqrt{Z_0}), \quad b = (V - Z_0 I) / (2\sqrt{Z_0}), \quad P = \tfrac{1}{2}(|a|^2 - |b|^2)$$

$$\Gamma = b/a = (Z - Z_0)/(Z + Z_0), \quad [b_1; b_2] = [S_{11}\ S_{12};\ S_{21}\ S_{22}]\,[a_1; a_2]$$

5.2 ABCD model and S-parameter derivation

A lossy line section of length ℓ, characteristic impedance Zc and complex propagation constant γ has

$$[ABCD]_{line} = [\ \cosh\gamma\ell,\ Z_c \sinh\gamma\ell;\ \sinh\gamma\ell / Z_c,\ \cosh\gamma\ell\ ]$$

$$\text{lossless case: } [\ \cos\theta,\ jZ_c \sin\theta;\ j \sin\theta / Z_c,\ \cos\theta\ ],\quad \theta = \beta\ell$$

Cascading sections multiplies their matrices. For a two-port with real reference impedances Z1 (port 1) and Z2 (port 2), with Δ = A·Z2 + B + C·Z1·Z2 + D·Z1:

$$S_{11} = (A Z_2 + B - C Z_1 Z_2 - D Z_1)/\Delta, \quad S_{21} = 2\sqrt{(Z_1 Z_2)}/\Delta$$

$$S_{12} = 2(AD - BC)\sqrt{(Z_1 Z_2)}/\Delta, \quad S_{22} = (-A Z_2 + B - C Z_1 Z_2 + D Z_1)/\Delta$$

For Z1 = Z2 = Z0 these reduce to the familiar forms S21 = 2/(A + B/Z0 + C Z0 + D). **Network built:** port 1 (WR-90, Z0) → line section 1 (ℓ1 = 0.5 m, γ = αc + jβ) → λg/4 transformer (Zt, ℓt) → load ZL = 0.6 Z0 (port 2). The load is the antenna feed: a reduced-height guide b_L = 6.096 mm (b_L/b = 0.6), so ZL = 0.6 Z0 = 266.1 Ω (equivalent).

The equivalent circuits and the cascade are drawn in Figure 2.

5.3 Impedance behaviour before matching

$$\Gamma_L = (0.6 - 1)/(0.6 + 1) = -0.25;\quad VSWR = (1 + 0.25)/(1 - 0.25) = 1.667;\quad RL = -20\log_{10}(0.25) = 12.04\ \text{dB}$$

$$\text{Mismatch loss} = -10\log_{10}(1 - 0.0625) = 0.280\ \text{dB};\quad |S_{21}| = 2\sqrt{0.6}/1.6 = 0.968\ \text{(lossless step)}$$

A 12 dB return loss is not acceptable (the specification requires ≥ 20 dB, VSWR ≤ 1.22): the match is **rejected**.

5.4 Matching: quarter-wave transformer

The load is real (ratio 0.6), so a single λg/4 section is the simplest compensator. In the equivalent-line model its characteristic impedance must be $Zt = \sqrt{(Z0 \cdot ZL)} = \sqrt{0.6}\, Z0 = 0.7746\, Z0 = 343.5\ \Omega$. Because $Z0 \propto b$, the section is a reduced-height guide of height

$$bt = \sqrt{(b \cdot b_L)} = 0.7746 \times 10.16\ \text{mm} = 7.87\ \text{mm}\ , \quad \ell t = \lambda g(10\ \text{GHz})/4 = 39.71/4 = 9.93\ \text{mm}$$

$$Zin = Zt^2/ZL = (0.7746\, Z0)^2 / (0.6\, Z0) = Z0 \ \rightarrow\ \Gamma = 0 \text{ at } 10\ \text{GHz}$$

Worked hand calculation of S-parameters at 10 GHz (lossless transformer)

$$[ABCD]t = [0,\ jZt\ ;\ j/Zt,\ 0]\ , \quad \Delta = jZt + j\, Z1Z2/Zt = 2jZt \ \ (\text{since } Zt^2 = Z1Z2)$$

$$S11 = (jZt - jZ1Z2/Zt)/\Delta = (Zt^2 - Z1Z2)/(Zt^2 + Z1Z2) = 0\ , \quad S21 = 2\sqrt{(Z1Z2)}/\Delta = 2Zt/(2jZt) = -j \ \rightarrow$$

$$|S21| = 1\ ,\ \angle S21 = -90°$$

$$RL = \infty\ , \quad IL = -20 \log|S21| = 0\ \text{dB (before conductor loss)}$$

Including conductor loss and band edges the computed values are in Table 8 and Figure 5. The transformer is a narrow-band device, but with ratio 0.6 the reflection stays below −29 dB across the entire 9.5–10.5 GHz band and below −20 dB from about 8.6 to 11.5 GHz.

5.5 Network summary and S-parameter plots

Parameter	Before matching	After matching	Target	Interpretation
Characteristic impedance Z0 (equivalent)	Zpv = 443.5 Ω; load 266.1 Ω	Zt = 343.5 Ω transformer; load 266.1 Ω	Reference = guide	Step ratio 0.6 is corrected by the 0.775 section
Reflection coefficient / VSWR	|Γ| = 0.246 / 1.65 (flat in band)	|Γ| = 4.4×10^{-5} / 1.00 at 10 GHz; ≤ 0.036 / ≤ 1.07 in band	VSWR ≤ 1.22	Match achieved, residual from λg(f) dispersion
Return loss (from S11)	12.2 dB	87 dB at 10 GHz; ≥ 29.0 dB over 9.5–10.5 GHz	≥ 20 dB	Passes with 9 dB margin
Insertion loss (from S21)	0.343 dB (silver guide) / 0.435 dB (initial)	0.064 dB at 10 GHz; ≤ 0.072 dB in band (line + transformer)	≤ 0.30 dB total	Total budget 0.16 dB incl. transition
Reciprocity / lossless check	S12 = S21; |S21−S12| ≤ 10^{-15}	S12 = S21; lossless (σ→∞): $\|S11\|^2+\|S21\|^2-1 = 1\times10^{-13}$; lossy: 1.46 % dissipated at 10 GHz	S12 = S21; $\|S11\|^2+\|S21\|^2 = 1$ (lossless)	Passive reciprocal network; S11 ≠ S22 (ports have unequal references)

Table 8: Network summary.

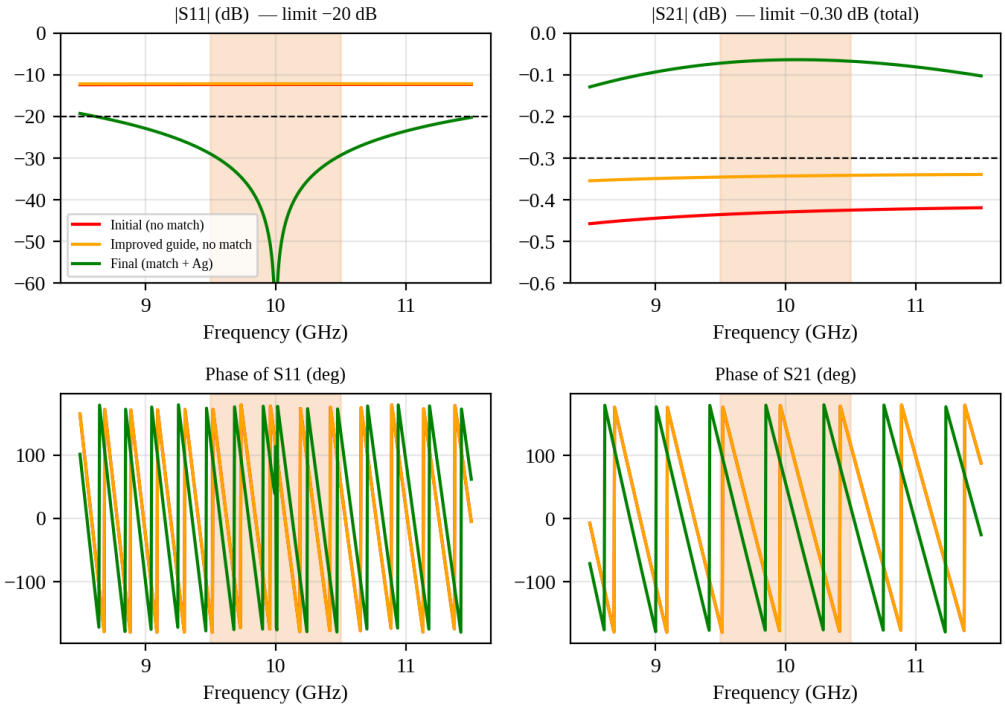


Figure 5: S-parameter magnitude and phase over 8.5–11.5 GHz (operating band shaded). Phase includes the 0.5 m line, hence the repeated wrap. The unmatched guide has $|S11| \approx -12.2$ dB; the matched design is below −29 dB in band.

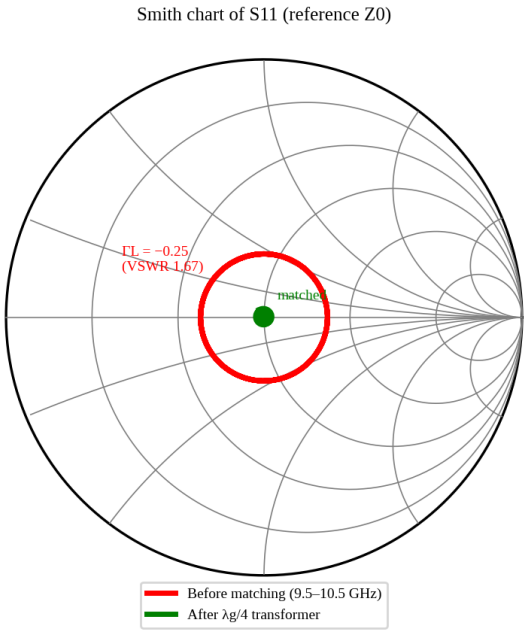


Figure 6: Smith-chart trajectory of S11 over 9.5–10.5 GHz, before and after matching.

5.6 Properties of the S-matrix

- **Reciprocity:** S12 = S21 to 10^{-15} (numerical round-off) – the waveguide and transformer contain only isotropic, passive material.
- **Symmetry:** S11 ≠ S22 because port 2 is referenced to a different impedance (ZL = 0.6 Z0); the network is not geometrically symmetric. For the line alone S11 = S22.
- **Losslessness:** with σ → ∞ the unitary condition $|S11|^2 + |S21|^2 = 1$ holds to 10^{-13}. With real metals the deficit $1 - |S11|^2 - |S21|^2 = 0.0146$ at 10 GHz equals the dissipated power; 10

log10(1 − 0.0146) = −0.064 dB, which is the IL computed from S21 – the S-parameters correctly reveal the conductor loss.

- **IL and RL from S:** IL = −20 log|S21| and RL = −20 log|S11|; mismatch loss is IL of the lossless case = −10 log(1 − |S11|²).

5.7 Independent verification

- **Route 1 (ABCD → S):** matrix cascade in NumPy (Appendix A).
- **Route 2 (impedance transformation):** Zin computed by two successive complex transformations Zin = Zc (ZL + Zc tanh γℓ)/(Zc + ZL tanh γℓ), giving Zin/Z0 = 0.99996 + j0.00008 and |Γ| = 4.4008×10⁻⁵, identical to the ABCD result (4.4008×10⁻⁵). Closed-form hand results (Γ = −0.25, RL 12.04 dB, ML 0.280 dB, |S21| = 0.968) agree with the code.
- **Route 3 (conservation / bandwidth):** unitary check above; single-section fractional bandwidth for Γm = 0.1 is about 50 % (Pozar formula), consistent with RL ≥ 20 dB from about 8.6 to 11.5 GHz (≈ 29 %) after allowing for waveguide dispersion.

A Smith-chart plot was produced (Figure 6). A full-wave 3-D simulation (CST/HFSS) of the two E-plane steps and the coax-to-guide transition is the recommended next verification (it was not run here); see 6.3 for the estimated influence of step discontinuities.

STAGE 6 – Validate, Critique and Defend the Final Design

6.1 Validation of each requirement by an independent method

- **Cutoff and mode range:** standard WR-90 datasheet (fc = 6.557 GHz, band 8.2–12.4 GHz, single-mode up to 13.1 GHz) agrees with c/2a.
- **Attenuation:** WR-90 copper datasheet loss at 10 GHz is about 0.11 dB/m; the formula gives 0.108 dB/m (smooth copper) – within 2 %.
- **Impedance and VSWR:** Smith chart (Figure 6) and the closed form $\Gamma = (ZL - Z0)/(ZL + Z0)$.
- **Breakdown:** standard formula for TE10 with E_br = 3 MV/m (1.05 MW) agrees with published WR-90 peak-power ratings of the order of 1 MW at sea level.

6.2 Critique of the assumptions

Assumption	Effect on results	Judgement
Perfect conductor for fields	Real walls add αc by perturbation (included); field distortion < δ/b ≈ 10^{-4}	Valid
Uniform lossless air dielectric	αd ≈ 4×10^{-5} dB/m; humidity and pressure change εr by < 0.06 %	Valid
Frequency-independent εr	Air: exact; for planar lines it would matter (dispersion ≈ 2 %)	Valid for the chosen structure
Neglected discontinuities (steps, flange gaps, transition)	Step susceptance estimate: B/Y ≈ 0.066 and 0.051 → RL degrades from 29.0 to 26.2 dB; retuning ℓt to 9.63 mm (−3 %) restores 28.9 dB	Moderate; fix by tuning (see 6.5)
Reference impedance via Z_PV	Only ratios enter S; result independent of V/I definition	Valid
Ideal load ZL = 0.6 Z0 (real, flat)	A real antenna has reactance and slope; will shift the optimum ℓt and limit RL	Must be re-measured after antenna integration

Table 9: Critique of assumptions.

6.3 Tolerance and frequency robustness

A grid of all combinations of ±t on a, b, the transformer height bt and the transformer length ℓt (81 corner cases) was evaluated at 41 frequency points across 9.5–10.5 GHz (the load height is held at its nominal value):

Tolerance class on a, b, bt, ℓt	Worst RL (dB)	Worst VSWR	Worst IL of line + transformer (dB)	Meets RL ≥ 20 dB?
±1 %	26.6	1.10	0.078	Yes
±2 %	24.1	1.13	0.088	Yes
±5 % (stress)	18.6	1.27	0.138	No (1.4 dB short)

Cutoff and band margins at ±5 % on a: fc10 = 6.24–6.90 GHz and fc20 = 12.49–13.80 GHz, so the worst margins are 9.5/6.90 = 1.376 (37.6 % above) and 10.5/12.49 = 0.841 (15.9 % below) – single-mode operation is preserved. Total attenuation at ±5 %: 0.138 (line + transformer, includes conductor change) + 0.09 (transition/flanges) + 0.0005 = 0.229 dB ≤ 0.30 dB ✓. The αc itself changes only −3 % / +6 % (0.121 dB/m for a +5 %, b −5 %; 0.132 dB/m for a −5 %, b +5 %; nominal 0.125 dB/m). **Conclusion:** the design is robust against realistic machining (±0.5 % is typical for WR-90 tubing) and satisfies RL at ±2 %; a ±5 % stress breaks the RL limit, so the transformer tolerance must be specified at ±2 % (±0.15 mm on bt, ±0.2 mm on ℓt).

6.4 EMC and safety

- **Shielding:** a 1.27 mm aluminium wall has absorption loss 8.686 t/δs = 8.686 × 1.27 mm / 1.0 µm ≈ 11 000 dB, so leakage is controlled only by seams. A gasketed or choke flange (design basis ≥ 100 dB) with conductive gaskets and flange bolts at a pitch of about 25 mm or less

meets the 90 dB specification. Any ventilation or test hole is built as a below-cutoff tube: a 3 mm hole, 10 mm deep, attenuates 32 t/d = 107 dB.

- **Out-of-band rejection (natural high-pass):** at 5 GHz (Wi-Fi/C2) the guide is below cutoff with $\alpha = (2\pi fc/c)\sqrt{(1 - (f/fc)^2)} = 137.4 \times 0.647 = 88.8$ Np/m = 772 dB/m, i.e. 386 dB over 0.5 m; at 2.4 GHz it is even higher. GNSS and C2 interference therefore cannot enter via the guide; 2nd-harmonic (20 GHz) must be handled by the filter of 3.5.
- **Grounding and connector choice:** bond the guide to the airframe ground at both ends through the flanges (low-impedance 360° bond), use a 2.92 mm / K connector for the PA (mode-free to 40 GHz) and keep the coax-to-guide probe on the centre line.
- **Exposure:** with 50 W average and leakage ≥ 90 dB the leaked power is < 0.5 µW (peak) – safe; antenna exposure zones are defined by the antenna gain and are outside the scope of the feed.

Basic EMC/safety test plan: (1) near-field probe scan of flanges and the transition at 500 W peak (leakage ≤ −90 dBc); (2) VNA S11/S21 at the flanges before and after thermal cycling; (3) conducted/radiated emission check of the 2nd harmonic at 20 GHz; (4) susceptibility with a 5 GHz and 2.4 GHz interferer injected next to the feed; (5) measure RF burn-in temperature (≤ 20 K rise at 50 W average).

6.5 Improvement and experimental test plan

- **Most serious limitation:** return loss at ±5 % tolerance (18.6 dB) and step discontinuities (RL 26 dB untuned). **Improvement:** (i) specify ±2 % on the transformer, (ii) shorten ℓt to 9.63 mm to compensate the step susceptance (restores 28.9 dB), (iii) if a wider band is needed, use a two-section binomial transformer (heights 8.94 mm and 6.93 mm, each $\lambda g/4$). This gives RL ≥ 20 dB over at least 8.5–11.5 GHz and 46 dB in band at nominal dimensions, but it does not remove the tolerance sensitivity (23.7 dB at ±2 %, 16.4 dB at ±5 %), so tolerance control remains the primary fix.
- **Test plan:** measure with a VNA (WR-90 TRL calibration) S11, S21 (9.5–10.5 GHz); confirm IL ≤ 0.30 dB and RL ≥ 20 dB; time-domain gating to locate the transition and steps; high-power test at 500 W peak / 50 W average with altitude (reduced pressure) chamber; CST/HFSS full-wave simulation of steps and the probe transition for comparison.

6.6 Specification verification table

Requirement	Specified	Initial design	Final design (worst case)	Satisfied?
Operating frequency / single-mode range	9.5–10.5 GHz inside TE10-only range; margins ≥ 25 % / ≥ 15 %	6.557–13.114 GHz; 44.9 % / 19.9 %	Same nominal; ±5 % a: 37.6 % / 15.9 %	Yes
Dominant mode and cutoff	TE10, fc ≤ 0.75·fmin = 7.1 GHz	TE10, 6.557 GHz	TE10, ≤ 6.90 GHz (a +5 %…−5 %)	Yes
Insertion loss / total attenuation	≤ 0.30 dB over 0.5 m	0.551 dB (1.10 dB/m)	0.160 dB nominal; 0.229 dB at ±5 %	Initial: No; Final: Yes
Impedance / VSWR	50 Ω interface; VSWR ≤ 1.22 (Zpv = 443.5 Ω)	VSWR 1.65	VSWR 1.07 nominal; 1.13 at ±2 %; 1.27 at ±5 %	Initial: No; Final: Yes (±2 %)
Return loss (S11)	≥ 20 dB	12.2 dB	29.0 dB nominal; 24.1 dB at ±2 %; 18.6 dB at ±5 %	Initial: No; Final: Yes (±2 %)
EMI shielding / isolation	≥ 90 dB; out-of-band ≥ 60 dB	Plain flange: not assured (seam leakage)	Gasketed/choke flange ≥ 100 dB (design basis, to be measured); ≥ 386 dB below 6.5 GHz by cutoff	Yes (design basis, to be verified)
Power handling / size	500 W peak, 50 W avg; mass ≤ 250 g; ≤ 30 × 15 mm	1.05 MW; ≈ 0.20 kg (0.12 kg guide + ≈ 0.08 kg flanges, est.)	1.05 MW (≥ 20 dB margin after derating); ≈ 0.21 kg; 25.4 × 12.7 mm	Yes

Table 10: Specification verification table.

FINAL ENGINEERING DECISION

SHOULD THIS STRUCTURE AND MODE BE ADOPTED FOR THE APPLICATION?

(i) Initial design (aluminium, rough bore, no matching): NO. Insertion loss 0.551 dB exceeds 0.30 dB, return loss 12.2 dB (VSWR 1.65) is far below 20 dB, and flange shielding is unproven. Mode behaviour is fine (TE10 only, margins 44.9 % / 19.9 %), but the dominant loss is mismatch (50.9 %), which the design does not address.

(ii) Final design (silver-plated polished WR-90, λg/4 transformer, optimised transition and gasketed flange): YES, subject to a ±2 % tolerance class on the transformer. Evidence from Stages 1–6:

- **Compliance:** loss 0.160 dB (0.229 dB at ±5 %) < 0.30 dB; RL 29.0 dB nominal, 24.1 dB at ±2 % > 20 dB; shielding ≥ 100 dB by design; mass ≈ 0.21 kg < 0.25 kg.
- **Dominant loss:** mismatch cut from 0.280 to 0.0055 dB, conductor from 0.149 to 0.064 dB; the transition (56 % of 0.16 dB) is now dominant.
- **Single-mode margin:** 44.9 % above fc10 and 19.9 % below fc20 (37.6 % / 15.9 % at ±5 %), with 1.05 MW breakdown (≥ 33 dB margin).
- **Matching achieved:** RL 12.2 → ≥ 29.0 dB; reciprocity and passivity verified; independent methods agree to 5 digits.
- **EMC and tolerance:** natural high-pass rejection (386 dB at 5 GHz over 0.5 m), seam-controlled shielding, RL robust to ±2 %.

If the answer is NO (remaining non-compliance and further change)

The only non-compliance is the ±5 % stress case (RL 18.6 dB, VSWR 1.27), and the uncertainty in the real antenna impedance and steps. **Further change:** (a) tighten the transformer tolerance to ±2 % or tune ℓt to 9.63 mm to cancel the step susceptance (RL 26.2 → 28.9 dB); (b) use a two-section binomial transformer (8.94 mm and 6.93 mm) if a wider band is required: nominal in-band RL rises from 29 dB to 46 dB with unchanged loss (0.068 dB) and unchanged mode behaviour (symmetrical steps, TE20 evanescent), but it does not fix the tolerance sensitivity, so (a) is still required. If the carrier moved below about 6 GHz a change of structure to semi-rigid coax would be required (Stage 2 sensitivity).

Appendix A – Code (Python/NumPy)

The script below reproduces the numerical results (worst-case RL and IL over 9.5–10.5 GHz). It is the code behind Figures 4–6 and Tables 7–8.

```
# WR-90 X-band feed: propagation, loss and ABCD/S-parameter model  (NumPy only)
import numpy as np
c=299792458.0; mu0=4e-7*np.pi; eta0=376.730313
a,b,L,f0=22.86e-3,10.16e-3,0.5,10e9
rL=0.6                          # load height / main guide height
NP=8.685889638                  # Np -> dB


def beta(f):  return np.sqrt((2*np.pi*f/c)**2-(np.pi/a)**2+0j)
def Zte(f):   return eta0/np.sqrt(1-(c/(2*a*f))**2)
def Rs(f,s):  return np.sqrt(np.pi*f*mu0/s)
def delta(f,s): return 1/np.sqrt(np.pi*f*mu0*s)
def K(f,s,r): return 1+2/np.pi*np.arctan(1.4*(r/delta(f,s))**2) if r>0 else 1.0
def alpha_c(f,s,r,b_=b):        # TE10 conductor attenuation, Np/m
    x=c/(2*a*f)
    return K(f,s,r)*Rs(f,s)/(b_*eta0*np.sqrt(1-x**2))*(1+2*b_/a*x**2)


def abcd_line(Zc,gam,l):
    ch,sh=np.cosh(gam*l),np.sinh(gam*l)
    return np.array([[ch,Zc*sh],[sh/Zc,ch]])


def s_params(M,Z1,Z2):          # real, unequal reference impedances
    A,B,C,D=M[0,0],M[0,1],M[1,0],M[1,1]
    Dn=A*Z2+B+C*Z1*Z2+D*Z1
    return ((A*Z2+B-C*Z1*Z2-D*Z1)/Dn, 2*np.sqrt(Z1*Z2)/Dn,
            2*(A*D-B*C)*np.sqrt(Z1*Z2)/Dn, (-A*Z2+B-C*Z1*Z2+D*Z1)/Dn)


def network(f,sigma,rough,matched):
    Z0=2*b/a*Zte(f)             # power-voltage equivalent impedance
    ZL=rL*Z0
    M=abcd_line(Z0,alpha_c(f,sigma,rough)+1j*beta(f),L)
    if matched:
        bt=np.sqrt(rL)*b; Zt=2*bt/a*Zte(f)
        lt=(2*np.pi/beta(f0)).real/4     # lambda_g/4 at 10 GHz
        M=M@abcd_line(Zt,alpha_c(f,sigma,rough,bt)+1j*beta(f),lt)
    return s_params(M,Z0,ZL)


f=np.linspace(9.5e9,10.5e9,101)
for tag,sig,rg,m in [("initial (Al, 1.5um, no match)",2.5e7,1.5e-6,False),
                     ("final   (Ag, 0.3um, lambda/4)",6.17e7,0.3e-6,True)]:
    S=np.array([network(x,sig,rg,m) for x in f])
    RL=-20*np.log10(abs(S[:,0])); IL=-20*np.log10(abs(S[:,1]))
    print(tag,"worst RL = %.2f dB, worst IL = %.3f dB, |S12-S21|max = %.1e"
          %(RL.min(),IL.max(),abs(S[:,1]-S[:,2]).max()))
```

Output of the script: initial design worst RL 12.33 dB, worst IL 0.435 dB; final design worst RL 28.97 dB, worst IL 0.072 dB, $|S12 - S21| < 10^{-15}$.

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